HARSHITA POOJARY

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TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, Go, Java, Scala, HTML/CSS
Frameworks & Libraries: Node.js, LangChain, PyTorch, TensorFlow, Keras, OpenCV, PySpark, AngularJS, Streamlit
Data Engineering Databases: Databricks, PySpark, Hadoop, MongoDB, Redis, MySQL, Cassandra, Elasticsearch, Milvus, DynamoDB
DevOps & MLOps: AWS, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, gRPC, REST APIs
Monitoring & Tools: Grafana, Prometheus, Kibana, OpenTelemetry, Git, Jira, Postman, Bruno, Swagger, PowerBI, Tableau

WORK EXPERIENCE

Software Engineer **Jun 2026 – Present**
CVS Pharmacy (Remote) *USA*

- Engineered an automated **Databricks** pipeline for standard-template rebate requests, reducing downstream final-processing runs by **30%** through automated rate generation, quality checks, and queue-based orchestration.
- Resolved 10+ production defects and cut debugging turnaround by [5%] across formulary ingestion, rate generation, and forecast review by tracing failures through **Azure Application Insights (KQL)**, **Databricks** job logs, and **MySQL** query plans, and adding unit, functional, and regression test coverage to prevent recurrences.

AI Engineer **Jul 2025 – Jun 2026**
Deep Defense Solutions (Remote) *USA*

- Built Agentic AI pipeline architecture using **Python**, **LangChain**, REST APIs, and **Elasticsearch** vector stores to identify false incident reports, automate routing, and reduce manual verification load for internal operations teams.
- Built guardrail-aware **LLM workflows** with finetuned ML and VLM models to classify incident reports by severity, credibility, and escalation risk, adding confidence scoring and fallback paths to prevent low-confidence AI outputs from triggering automated decisions.

Machine Learning Engineer **Dec 2021 – Jul 2023**
Reliance Jio *Mumbai, India*

- Reduced face registration latency by **40%** by deploying RetinaFace models with **TensorRT FP16** compression using **Python** and **C++** on **T4/A100 GPUs**, enabling real-time **computer vision** inference across 2 distributed **Kubernetes** clusters.
- Increased deployment velocity by **50%** by automating **CI/CD** pipelines for Linux-based backend services using **Docker, Kubernetes, GitLab,** and **Azure**, improving developer process and release reliability with ownership of the operational health.
- Improved production reliability by **20%** by instrumenting logs, metrics, and distributed traces with **OpenTelemetry**, reducing MTTR for large-scale distributed service failures.
- Implemented **spoof-detection** across **20+ sites** at **90%** accuracy by curating **in-house annotated datasets** tailored to deployment scenarios, fine-tuning **ResNet-32** classifiers, and integrating FastAPI endpoints with real-time drift detection and running A/B tests.
- Mentored **4 interns** through dataset curation, model validation, and code and design reviews, improving reproducibility, code quality, and production readiness.

Software Engineer **Jun 2019 – Dec 2021**
Reliance Jio *Mumbai, India*

- Architected Node.js microservices and content-delivery REST APIs for consumer or business facing products serving 100M+ users on Kubernetes with **MongoDB, MySQL, Redis,** and **Elasticsearch**, implementing sharded data flows, RBAC-based secure access control.
- Built large-scale distributed ingestion pipelines for real-time data processing with **Kafka**, **Node.js, Go, SQL, Kubernetes**, improving throughput **35%** across multi-device platforms with high availability.
- Implemented **gRPC-based authorization** between content microservices and a centralized **UserService**, validating user access via **OAuth/RBAC** and securing inter-service communication.
- Reduced API downtime by **15%** by improving performance of **RESTful APIs** and **Kubernetes** services with integration/regression tests, CI validation, and observability dashboards with **Grafana**, **Prometheus**, and **Kibana**.
- Enhanced user retention by **12%** by developing recommendation infrastructure for personalized content ranking, leveraging user-behavior and geo-based features across 100M+ users with A/B testing.

Software Engineer **Jan 2019 – Jun 2019**
TaksyKraft (Remote) *Hyderabad, India*

- Developed **Python** and **SQL** APIs for real-time engagement prediction, improving targeting accuracy and increasing client ROI by **10%**.
- Shortened reporting cycles by **40%** by building **ETL pipelines** and **Tableau dashboards**, accelerating enterprise decision-making.

PROJECTS

Meeting Copilot (FastAPI, WebSockets, LangGraph, ChromaDB, SQLite, Slack API, Gmail API)

- Designed and built a multi-tenant **Agentic AI** platform using **FastAPI**, **WebSockets**, **LangGraph**, **ChromaDB**, **SQLite**, **Slack API**, and **Gmail API** to process live meeting transcripts, extract verified action items, and deliver automated summaries across enterprise workflows.

Unified Semantic Space for Multimodal Retrieval (PyTorch, Python, LanceDB)

- Designed a multimodal retrieval framework aligning **BridgeTower** text–image embeddings for video-based Q&A, integrating **LanceDB** for semantic search with ranking and deploying an interactive **Streamlit** interface.

EDUCATION

University of Southern California, Los Angeles, CA. **Aug 2023 - May 2025**
Master of Science - Computer Science **GPA: 3.81/4.00**

University of Mumbai, Mumbai, India. **Jul 2015 - May 2019**
Bachelor of Engineering - Computer Engineering **GPA: 3.83/4.00**